SpecVision AI

AI-Powered Product Specification and Cost Estimation for Custom Jewelry

What this is

An ML engineering system for the custom jewelry domain. It takes a short sales brief (and optionally a reference photo of a ring) and produces: a structured product specification, a ranked match against a 42-item inventory catalog, a classical computer-vision estimate of visible gemstones in the photo, an indicative price, and an explicit confidence/review summary. The core design goal is honesty about uncertainty: every field carries a source (stated by the user / inferred by the model / unknown) and a confidence level, and the system is built to say "unknown" or "needs review" rather than present a guess as a confident fact.

What was built

- Backend (FastAPI, Python): LLM-based extraction (Groq, live mode) with a fully functional deterministic offline fallback -- no API key required to run the whole pipeline.
- Deterministic inventory matching: a weighted attribute-scoring engine (no LLM) that ranks the catalog, with a documented 0.72 auto-selection threshold below which it explicitly declines to claim a reliable match.
- Classical computer vision (OpenCV, no paid vision API): Hough-circle + contour hybrid pipeline that estimates visible gemstone count, with its own confidence assessment and image-quality warnings.
- Pricing engine: indicative price derived from the matched catalog item plus documented adjustments, or explicitly "unavailable" when there is nothing reliable to price from.
- Confidence & review layer: cross-checks the CV estimate against the brief, runs deterministic plausibility checks (implausible carat, metal/purity contradictions), and raises stable, actionable review flags instead of hiding uncertainty.
- React/TypeScript frontend consuming the real API contract (no mocked data), with distinct live-mode and offline-mode indicators.
- A 16-case labeled evaluation harness using real photographs (not synthetic), a 52-test automated pytest suite, and a documented failure analysis of the three worst observed cases with root causes and proposed fixes (EVALUATION.md).

How to run it

cd backend && .venv\Scripts\Activate.ps1 && uvicorn app.main:app --reload --port 8000
cd frontend && npm run dev (opens at http://localhost:5173)

Full setup instructions, environment variables, and architecture are in README.md. Measured evaluation metrics and failure analysis are in EVALUATION.md.

Test case 1 -- successful end-to-end match

Brief: "A solitaire ring in 18k yellow gold with a round diamond center stone, about 0.45 carats, tension setting, eight moissanite side stones, band width 1.7mm." (no photo)

Mode: offline/demo (deterministic, no API call -- fully reproducible). Run with:

python evaluation/runner/run_evaluation.py --mode offline

Extracted specification

Product type	ring	**HIGH**
Design category	solitaire	**HIGH**
Metal	18k yellow gold	**HIGH**
Setting type	tension	**HIGH**
Center stone	round diamond, 0.45ct	**HIGH**
Side stones	8x moissanite	**HIGH**
Band width	1.7mm	**HIGH**

Inventory match

Selected: SV-001 -- "Solitaire Round Yellow Gold Ring" -- match score 100% (threshold: 72%) -- PASSED, auto-selected.

Indicative price

USD 2,568.00 (base catalog price, no adjustments needed -- spec was within the matched item's carat/side-stone range). Formula: indicative_total = catalog_unit_price + extra_carat_adjustment(if spec carat exceeds catalog max) + extra_side_stone_adjustment(if spec side count exceeds catalog count)

Confidence & review

Status: AUTO APPROVED

No review flags were raised -- every stated attribute was extracted correctly and matched a real catalog item within tolerance.

Test case 2 -- honest reporting of a real CV limitation

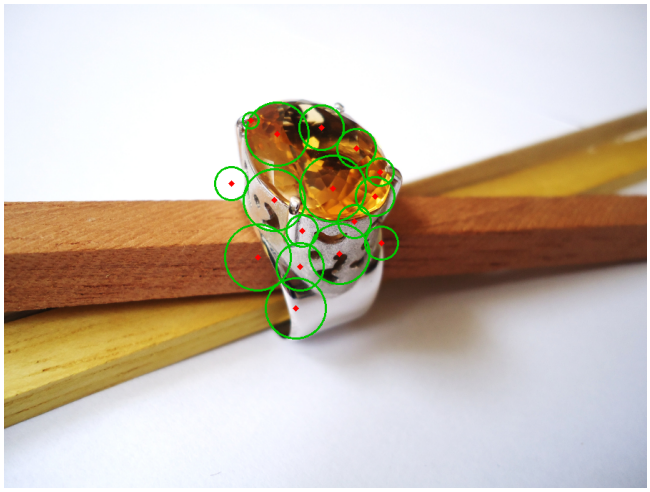
Brief: "A sterling silver ring with a large oval citrine centerpiece and an open filigree band, no side stones." (real photo, Wikimedia Commons CC BY-SA, see evaluation/images/PROVENANCE.md)

Mode: offline/demo. This exact case is analyzed in depth as "Failure #1" in EVALUATION.md.

Original photo



CV detections overlaid (green circles)



Extracted specification

Metal	925 sterling silver	HIGH
Center stone type	Unknown	LOW
Center stone shape	oval	HIGH
Side stones	0	HIGH

"Citrine" is not in the system's controlled gemstone vocabulary, so center_stone_type is correctly left Unknown rather than guessed -- a deliberate, documented scope limitation, not a crash.

Computer vision result

Visible components detected	16	HIGH

The photo's pierced filigree metalwork was mistaken for 16 small gemstones by the classical detector, with HIGH self-reported confidence and no image-quality warning -- the ring actually has zero side stones. This is a real, known limitation of the classical CV baseline (filigree/pierced metalwork resembles a ring of small stones), documented in EVALUATION.md as the project's #1 worst observed failure, with a proposed fix (geometric-regularity check).

Why the system still catches the error

Because the brief explicitly states "no side stones," the confidence layer cross-checks the CV count against the brief and finds a disagreement ratio far above the 40% threshold -- so this case is correctly flagged for human review despite the CV module's own overconfidence. If the same photo were paired with a brief that simply didn't mention side-stone count, this cross-check would not fire -- a limitation stated explicitly in EVALUATION.md, not hidden.

Confidence & review

Status: NEEDS REVIEW

[WARNING] CV_BRIEF_COUNT_DISAGREEMENT

CV detected 16 visible component(s) vs. 0 implied by the brief (relative difference 1600%).

[WARNING] INCOMPLETE_SPEC_FOR_PRICING

Specification is missing or uncertain in fields that affect pricing: center_stone.gemstone_type, center_stone.carat.

[WARNING] NO_RELIABLE_INVENTORY_MATCH

No candidate reached the auto-selection threshold of 0.72 (best score: 0.61). A high similarity score does not prove exact product

equivalence; results below threshold are shown for reference only and require human review.

[WARNING] PRICE_UNAVAILABLE

No inventory candidate met the auto-selection matching threshold, so there is no defensible catalog reference to price from. An indicative price would be a guess.

Test case 3 -- real photo, successful match and price

Brief: "An 18k yellow gold solitaire ring with a round diamond center stone and no side stones." (real photo, Wikimedia Commons CC BY-SA, see evaluation/images/PROVENANCE.md)

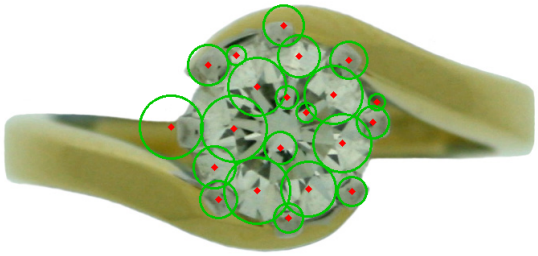
Mode: offline/demo (deterministic, no API call). Run with:

python evaluation/runner/run_evaluation.py --mode offline

Original photo



CV detections overlaid (green circles)



Extracted specification

Product type	ring	HIGH
Design category	solitaire	HIGH
Metal	18k yellow gold	HIGH
Center stone	round diamond	HIGH
Side stones	0	HIGH

Inventory match

Selected: SV-001 -- "Solitaire Round Yellow Gold Ring" -- match score 81% (threshold: 72%) -- PASSED, auto-selected.

Indicative price

USD 2,568.00 -- computed from the matched catalog item.

Computer vision result

Visible components detected 21 MEDIUM

This is a clean studio product photo of a plain solitaire with no side stones, but the band's polished-gold highlight still produces enough false-positive detections to push confidence down to MEDIUM. Specification extraction, inventory matching, and pricing all succeeded independently of this -- the three subsystems are deliberately decoupled, so a CV caveat on one image does not block the rest of the pipeline from producing a usable, priced result.

Confidence & review

Status: NEEDS REVIEW

Flagged for review -- not because the match or price are unreliable, but because the CV module honestly disagrees with the brief's claim of zero side stones. That disagreement is surfaced explicitly below rather than silently resolved either way.

[WARNING] CV_BRIEF_COUNT_DISAGREEMENT

CV detected 21 visible component(s) vs. 0 implied by the brief (relative difference 2100%).

[WARNING] INCOMPLETE_SPEC_FOR_PRICING

Specification is missing or uncertain in fields that affect pricing: center_stone.carat.

[WARNING] IMAGE_QUALITY_INSUFFICIENT

Image quality or detection stability is insufficient for an automatic visible-stone count.

Measured evaluation results (16 labeled cases)

Full methodology, per-field breakdowns, and the three-worst-failure analysis are in EVALUATION.md. Figures below are read directly from evaluation/results/evaluation_offline.json -- reproducible with: python evaluation/runner/run_evaluation.py --mode offline

Extraction accuracy (offline / deterministic mode)

- Categorical fields (metal type, purity, color, setting, stone type/shape): 92-100% exact-match accuracy on cases with a known gold value; design_category 71% (documented taxonomy-overlap limitation).
- 0 fabricated values across every field where the correct answer was "unknown."

Inventory matching

- Top-1 / Top-3 hit rate: 100% on the one case engineered to match a specific catalog item.
- No-reliable-match handling: 100% correct (8/8 cases engineered to match nothing reliably).

Computer vision

- Mean absolute error: 16.71 stones across 7 real photos -- driven almost entirely by two explainable, documented failure modes (filigree metalwork and faceted-stone reflections), not a uniform small error. See EVALUATION.md for the per-image breakdown.

Review behavior

- Precision: 93% | Recall: 100% | Flag rate: 88% (14/16 cases correctly got at least one actionable flag).

Operational

- 52/52 automated tests passing. 16/16 evaluation cases executed successfully, 0 crashes.
- Live-mode run: 15 real Groq API calls, $0.0000 cost (free tier), including one real HTTP 429 rate-limit recovery and one real malformed-JSON repair -- both handled correctly.